SLIIT UNI
THE KNOWLEDGE UNIVERSITY

IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-226

1. Topic (12 words max)

Hospital Resource Optimization and Demand Forecasting Using Temporal Analytics, NLP, Explainable AI

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Efficient resource management in large-scale hospitals remains a persistent challenge, especially during seasonal illness surges, public holidays, or emergencies. Traditional resource planning methods are often reactive and static, making it difficult to predict sudden spikes in patient admissions, doctor availability issues, or medicine shortages.

Hospitals frequently suffer from either underutilization or overutilization of resources—such as vacant beds in one ward while others are overcrowded. Similarly, patient appointments may be scheduled on days when specific doctors are unavailable, and medicine stockouts occur due to a lack of predictive planning. These issues collectively degrade the quality of care and operational efficiency. Most existing systems fail to integrate historical patterns or unstructured medical records (like doctor notes or discharge summaries) into a cohesive, predictive framework. Furthermore, a lack of explainability in AI-based systems limits trust among healthcare professionals.

This research proposes a comprehensive solution using AI to analyze historical data, predict future demand, and optimize hospital resource usage. By incorporating time-series models for bed and medicine demand forecasting, NLP for identifying illness trends, and explainable AI for transparency, the system enables hospitals to prepare proactively and make informed, data-driven decisions.

Scanned with CamScanner

References:

Rajkomar, A., Dean, J., & Kohane, I. (2019). Machine learning in medicine. *The New England Journal of Medicine*, 380(14), 1347–1358. https://doi.org/10.1056/NEJMra1814259

Choi, E., Bahadori, M.T., Schuetz, A., Stewart, W.F., & Sun, J. (2016). Doctor AI: Predicting clinical events via recurrent neural networks. *Journal of Machine Learning Research*, 18(1), 1–18.

Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., & Philip, S.Y. (2020). A comprehensive survey on graph neural networks. *IEEE Transactions on Neural Networks and Learning Systems*, 32(1), 4–24.

Miotto, R., Li, L., Kidd, B.A., & Dudley, J.T. (2016). Deep patient: An unsupervised representation to predict the future of patients from the electronic health records. *Scientific Reports*, 6, 26094.

Esteva, A., Robicquet, A., Ramsundar, B., Kuleshov, V., DePristo, M., Chou, K., ... & Dean, J. (2019). A guide to deep learning in healthcare. *Nature Medicine*, 25, 24–29. https://doi.org/10.1038/s41591-018-0316-z

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed system is a modular, AI-powered solution designed to optimize hospital resources proactively. It is structured around four main components:

Temporal Bed Usage Forecasting + Reallocation Engine
Utilizes LSTM or Transformer-based models to analyze historical patient inflow and bed occupancy data. It forecasts ward-wise bed demand and recommends dynamic reallocation to avoid both overcrowding and underutilization.

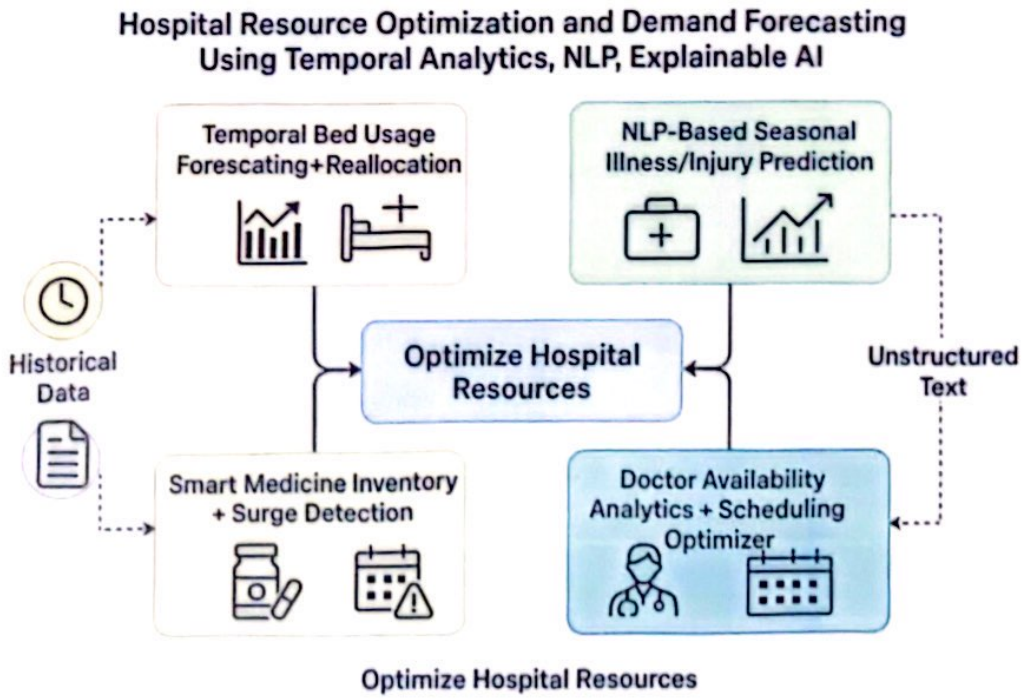
NLP-Based Seasonal Illness/Injury Prediction Module
Applies Natural Language Processing to unstructured data such as discharge summaries, diagnosis reports, and doctor notes. This identifies patterns in seasonal illnesses or event-driven injuries (e.g., flu outbreaks or firecracker accidents).

Doctor Availability Analytics + Scheduling Optimizer
Analyzes historical doctor attendance to predict likely absences. It flags high-risk days (e.g., frequent leaves on Fridays) and suggests optimized appointment scheduling to reduce patient-doctor mismatches.

Smart Medicine Inventory + Emergency Surge Detection
Forecasts medicine demand based on predicted patient cases and public events. It ensures timely restocking, especially for high-demand or emergency medications, using AI to predict potential accident surges.

Scanned with CamScanner

An **Explainable AI (XAI)** layer is integrated into the system to provide clear reasoning behind each prediction or recommendation. This enhances trust and usability among hospital staff.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

- This project combines expertise from multiple domains:
  Artificial Intelligence & Deep Learning:
  Essential for developing time-series forecasting models (e.g., LSTM, Transformer) that analyze trends in patient admissions, ward occupancy, and medicine usage over time.
  Natural Language Processing (NLP):
  Required for processing unstructured medical text such as discharge notes and diagnosis summaries to identify patterns in illnesses and injury types across seasons.
  Healthcare Systems and Operations Management:
  Understanding hospital workflows, resource limitations, staff scheduling dynamics, and the administrative structure is crucial for creating practical and effective recommendations.
  Explainable AI (XAI):
  Important to build models that can justify their outputs in human-understandable ways, making it easier for hospital staff to trust and act on AI suggestions.
  Data Engineering & Integration:
  Necessary to clean, preprocess, and manage heterogeneous data sources from hospital databases including structured logs and unstructured reports.
-

Scanned with CamScanner

Data Requirements:

- **Structured Data**:
  - 2–3 years of patient admission and discharge logs
  - Historical ward occupancy rates (bed usage per day/ward)
  - Doctor duty schedules and attendance records
  - Medicine inventory and restocking logs
- **Unstructured Data**:
  - Discharge summaries
  - Medical notes and diagnosis reports
  - Administrative communications (e.g., leave notices)
- **External Data**:
  - Public holiday/event calendars
  - Historical data on seasonal illnesses and local outbreaks

All patient and hospital data will be anonymized in compliance with ethical standards and privacy regulations.

8. Objectives and Novelty

Main Objective To develop an AI-driven system for proactive hospital resource forecasting using deep learning, NLP, and explainable AI			
Member Name with Registration No	Sub Objective	Tasks	Novelty
LOCHANA J L S IT22555526	Predict ward-level bed demand	Train time-series models on historical occupancy	Dynamic bed reallocation recommendations
HERATH H M M B IT22252890	Detect seasonal illness/injury trends	Apply NLP on medical records	Forecast future surges from unstructured text
ADIKARI A M C G IT22075758	Optimize doctor-patient scheduling	Analyze doctor availability patterns	Avoid mismatches using predictive analytics
WIJETHUNGA S R IT22179326	Forecast medicine requirements	Link events to demand spikes	Prevent stockouts during seasonal surges

Scanned with CamScanner

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
LOCHANA J L S IT22555526	Focuses on deep learning for temporal bed forecasting using LSTM/Transformer, aligning with time-series AI specialization.
HERATH H M M B IT22252890	Applies NLP techniques to medical records to uncover seasonal illness patterns, aligning with unstructured data analytics specialization.
ADIKARI A M C G IT22075758	Specializes in staff attendance analysis and patient-doctor scheduling optimization using predictive modeling.
WIJETHUNGA S R IT22179326	Develops AI models for forecasting medicine demand and emergency readiness, focusing on event-driven healthcare analytics.

Scanned with CamScanner

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Kanishka	Yapa	
Co-Supervisor	Mr.	Uditha	Dharmakeerthi	27/06
External Supervisor	Mrs.	Akshi	DeSilva	
Summary of external supervisor's (if any) experience and expertise				

Scanned with CamScanner

SLIIT UNI
THE KNOWLEDGE UNIVERSITY

IT4010 Research Project – 2025 July

Topic Assessment Form

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Need to get ethical clearance for patient information.

Scanned with CamScanner

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.

Scanned with CamScanner